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1. a). Show that $\phi : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p'(0)$ is linear functional. $\boxed{8m}$

b). Find a dual basis corresponding to $\{1, t, \cdots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{n} b_i x^i$ and $c \in \mathbb{R}$ be given.

Then,
$$\phi(cf+g) = (cf+g)'\big|_{t=0} = c \cdot f'\big|_{t=0} + g'\big|_{t=0} = c \cdot \phi(f) + \phi(g)$$
by the linearity of differentiation. This shows the a).

To show b), let
$$\beta^* := \left\{ \frac{1}{k!} \cdot \phi_k : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p^{(k)}(0) \;\middle|\; k = 0, 1, \cdots, n \right\}$$

Then, for each $k = 0, 1, \cdots, n$,
$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in β^* are all linear functional, so the β^* is dual basis.

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2. Let $A \in M_{m\times m}(\mathbb{R})$, $\boxed{15m}$: A^n 이라 얼탐, $n=2$ 인때부터 생각

a). For $n \in \mathbb{N}$, $\operatorname{rank} A^n \le \operatorname{rank} A$.

b). Let $L_A : \mathbb{R}^m \to \mathbb{R}^m ; x \mapsto Ax$.

Show that for $n \ge 2$,

$$\operatorname{rank} A^n = \operatorname{rank} A \text{ implies } \operatorname{Ker} L_A \cap \operatorname{Im} L_A = \{0\}.$$

Sol) a). Given $x \in \operatorname{Im} A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.
So, $x = A^n y = A^{n-1} A y \in \operatorname{Im} A$, for $n \ge 2$. Thus $\operatorname{Im} A^n \subseteq \operatorname{Im} A$, it follows $\operatorname{rank} A^n \le \operatorname{rank} A$. If $n=1$, Clear.

b). Given $x \in \operatorname{Ker} A$, $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$, so $x \in \operatorname{Ker} A^n$, so $\operatorname{Ker} A \subseteq \operatorname{Ker} A^n$.
Since $\operatorname{rank} A^n = \operatorname{rank} A$, the dimension theorem gives $\operatorname{nullity} A^n = \operatorname{nullity} A$.
Therefore, since $\mathbb{R}^m$ is finite dimensional, So is $\operatorname{Ker} A^n$, thus $\operatorname{Ker} A = \operatorname{Ker} A^n$.
Now, let $x \in \operatorname{Ker} L_A \cap \operatorname{Im} L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.
Then, $A^{n-1} x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} \cdot Ax = 0$, and this implies $y \in \operatorname{Ker} A^n$.
Since $\operatorname{Ker} A^n = \operatorname{Ker} A$, $Ay = 0$, so $x = Ay = 0$, so proved. ⊣

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3. Find all $a,b,c \in \mathbb{R}$ such that $\boxed{6m}$

$$A=\begin{bmatrix}2&0&0\\a&2&0\\b&c&-1\end{bmatrix}$$ is diagonalizable.

Sol). Since the Charatorstic polynomial of A is regardless of a,b,c,

$$\chi_A(t)=(t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the charatorstic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I)=\begin{bmatrix}0&0&0\\a&0&0\\b&c&-3\end{bmatrix}\cdot\begin{bmatrix}3&0&0\\a&3&0\\b&c&0\end{bmatrix}=0.$$

Calculating gives

$$=\begin{bmatrix}0&0&0\\3a&0&0\\ac&0&0\end{bmatrix}$$

therefore, $a=0$, $b\in\mathbb{R}$, $c\in\mathbb{R}$ are all possible (a,b,c), [2025. Fall,

$\boxed{20m}$

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that

$$\det A = \pm \det B.$$

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol) Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

and $\det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$

so, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. 10m.

Define a map

$$\varphi : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to \mathbb{R}^{\times} : (A, B) \mapsto (\det A \cdot \det B^{-1})^2$$

Then, $\ker \varphi = N$ and φ is homomorphism, onto $(0, \infty)$, because the det preserves multiplication, so is φ, and ontoness is given by: fix $B = I$, and $A = \operatorname{diag}(\sqrt{r}, 1, 1, \cdots, 1)$ where $r > 0$ is arbitrary.

By the first-isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times)$$

10m. φ

Comment: φ 을 먼저 찾고 시작하고, hom 임을 보이면 앞 계산 과정이 날라감!

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10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G.

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.
By the Cauchy's theorem, G has order 3 element x and order 7 element y.
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \mod 21 \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g, $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \mod n$ has a solution iff $\gcd(a, n) \mid b$.
And in such a case, it has $\gcd(a, n)$ solution

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6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$. Define $\boxed{22m}$

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\},$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n\to\infty} a_n = \sup\{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0,1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.
Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.
Let a be a limit point of $\{a_n\}^*$. Then, given $k \in \mathbb{N}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
For $k=1$, choose such a $p_1 \in B_1(a) \setminus \{a\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{1-|p-a|}(p)$, then $a_{p_1} \in B_1(a)$.

For chosen $a_{p_1}, \cdots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
Then, similarly, there is $a_{p_k} \in B_{\frac{1}{k}}(a)$ such that $p_k > p_{k-1}$,

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point a is contained in $\{a_n\}^*$,
so it is closed. ◻

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup\{a_n\}^* = \max\{a_n\}^* = a \text{ for some subsequential limit } a.$$

So, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < a + \varepsilon$ (since $a = \max\{a_n\}^*$)
and there exists infinitely many terms in $\{a_n\}$ such that $a - \varepsilon < a_n$ (since $a_{n_k} \to a$)
Thus $\limsup_{n\to\infty} a_n = a$.

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \cdots$$

Then, given $b \in [0,1]$, there exists a subsequence in $\{b_n\}$ converges to b, because: for any $\varepsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \varepsilon,$$

So we can construct the subsequence. ◻

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